INCH-POUND

A-A-52034A
February 13, 1992

COMMERCIAL ITEM DESCRIPTION

DRAFTING INSTRUMENT SETS

AND DRAFTING INSTRUMENTS

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GG-D-600D.

This specification covers four drafting instrument sets, and individual drafting instruments. The drafting instruments are intended for use by architects, engineers, scientists and draftsmen for general drafting practice.

Salient characteristics:

Classification. Drafting instruments shall be supplied in four classifications of sets (see table I):

Set 1--Field set
Set 2--Office set
Set 3--Pocket set
Set 4--Mini set

When required, individual drafting instruments shall be as specified. Each individual instrument shall be supplied with a plastic case with a hinged lid and fastener, size fitted to the instrument; or a plastic vinyl pouch with foldover tucked flap.

Beneficial comments (recommendations, additions, deletions) and any other data which may improve this document should be sent by letter to: Commander, US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A

FSC 6675

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TABLE I. Drafting instrument sets.

Description	Set 1 field	Set 2 office	Set 3 pocket	Set 4 mini
Case	1	1	1	1
Ruling pen, small	1	1		1
large	1	1	1	
Small Bow compass, pen	1	1		
pencil	1	1		
combination				1
Large Bow compass				1
Compass, large, friction	1	1	1	
Lengthening bar	1	1		
Dividers, small	1	1		
large	1	1		1
Center horn		1		
Spare parts and lead holder	1	1	1	1
Screw driver (when required)	1	1	1	1
Protractor, type I, class A, 8-Inch		1		
Triangle, class A, 30° -60° 6-inch		1		
class B, 45° 4-inch		1		
Ruling pen, contour		1		
railroad		1		
Curve, pattern 1		1		
pattern 2		1		

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Material. The instruments shall be corrosion-resisting steel, or chrome- and nickel-plated brass, at the contractor's option. Materials shall be free from defects which could impair the function or appearance of the instruments. Surfaces shall be smooth, free from burrs and sharp corners (except needles), and shall be polished or have a satin finish. The steel shall conform to UNS S30100, S30400, S42000, or S43000, condition and temper as required. Brass shall conform to ASTM B 36 copper alloy UNS C26800, temper H02, or ASTM B 455 copper alloy UNS CS85000, minimum Rockwell B Hardness of 42 and shall be chrome-plated to a thickness of not less than 0.0004 inch or nickel-plated to a thickness of not less than 0.0003 inch. Aluminum for handles, lengthening bars, and other parts shall conform to ASTM B 211, UNS A92011 or A92024.

Design. Instruments shall be constructed of solid material, truss type, or hollow tube. In truss type, each truss shall be not less than 0.060 inch thick. If hollow tube type, the wall thickness shall be not less than 0.045 inch throughout the length of the piece. All compass and divider points shall be aligned to register accurately in closed position. Attachments, such as lengthening bars or beams, and pen and pencil parts, shall fit freely; and when tightened, shall not twist or turn.

Compass and divider handles. Compass and divider handles shall be the same material as the instrument and shall be finely knurled or shall have fine longitudinal corrugations. Handles shall be between 0.625 and 0.875 inch long, and between 0.187 and 0.312 inch in diameter.

Screws. All screws and screw studs shall be corrosion-resisting steel. Screw heads and adjusting nuts shall be knurled and shall be leaded brass conforming to ASTM B 455, copper alloy UNS C38500, or corrosion-resisting steel.

Needle points. Needle points shall be nickel-plated or corrosion-resisting steel. Needles shall have a shoulder point on one end and a needle point on the other. Each point shall be concentric with the body of the needle, and shall be sharp.

Pencil leads. Each instrument requiring pencil leads shall have a pencil lead firmly clamped in its proper socket. Pencil leads shall be between 1 and 1.25 inches long and 0.078 inch, +0.002 inch or -0.005 inch thick. Pencil lead shall have a 4H degree of hardness and shall have a chisel point.

Cases. Each set of drafting instruments shall be contained in a fitted case with a hinged lid. The case shall be wood, plastic, or aluminum. The bottom of each case shall have recesses shaped to fit the instrument for which it is intended. When in the case, no instrument shall come in contact with any other instrument or

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attachment in the case. The inside of the case, including that part that covers the instruments when the case is closed, shall be lined with dark colored velvet, velveteen, or plastic foam rubber. Fasteners shall be provided to keep the case closed when not in use. Fasteners and hinges shall be corrosion-resisting metal.

Ruling pens (except contour and railroad pens). Pen blades shall be spring-blade, spring-hinge, or cross-hinge types. Blade points shall have a hardness of 47 ±7, on the Rockwell C-scale or equivalent. The blades shall be sharpened and set, and shall be adjustable to permit drawing clean, uniform, unbroken lines between 0.006 inch and 0.03-inch wide on drawing paper, tracing cloth, and on matte-surfaced plastic sheet. The points shall not cut into the drafting medium under normal use.

Ruling pen handles. Ruling pen handles shall be rigid plastic or metal, and shall be between 0.187 and 0.25 inches in diameter.

Small ruling pen. The small ruling pen shall be between 4.5 and 5.5 inches long. The blade portion shall be 2 inches, ±0.25 inch, long.

Large ruling pen. The large ruling pen shall be between 5.5 and 6.5 inches long. The blade portion shall be 2 inches, ±0.25 inch.

Bow compass. The action of the legs of the bow compass shall be controlled by a center adjusting thumbscrew. The adjusting screw shall open and close the spacing firmly and smoothly and shall hold the legs rigidly while drawing circles.

Small bow compass. The small bow compass shall be between 3.5 and 4.5 inches long, and shall permit drawing circles from 0.031 inch to at least 2 inches in diameter.

Compass pen. The compass pen shall be provided with a removable and adjustable needle in one leg and a ruling pen in the other leg. The pen blades shall be as specified in the paragraph on ruling pens, above, except that the maximum width of the line shall be 0.03 inch.

Compass pencil. The compass pen shall be provided with a removable and adjustable needle in one leg and a pencil-lead holder in the other leg. The pencil-lead holder shall accommodate the pencil leads as specified above.

Combination bow compass. The combination bow compass shall be provided with a removable and adjustable needle in one leg, and the other leg shall have an interchangeable pen holder and pencil holder. The pen holder or the pencil holder shall be held firmly in place without movement when tightened. The pen holder portion shall accommodate pen blades conforming to the paragraph on ruling pens

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(except contour and railroad pens), above. The pencil holder portion shall accommodate leads conforming to the paragraph on pencil leads, above.

Large bow compass. The large bow compass shall be between 6 and 6.5 inches long, and shall permit drawing circles from 0.062 inch to at least 5 inches in diameter. One leg of the compass shall be provided with a removable and adjustable needle, and the other leg shall have an interchangeable pen holder and pencil holder. The pen holder or the pencil holder shall be held firmly in place without movement when tightened. The pen holder portion shall accommodate pen blades conforming to ruling pens, above. The pencil holder portion shall accommodate leads conforming to the paragraph on pencil lead above.

Protractor. Protractors shall be type I semicircular, class A, 8-inch. Protractors shall be of brass or plastic. Numbers may be slanted in the direction of increase. The outer row of numbers on the graduated arc shall read clockwise and the inner row of numbers shall read counterclockwise.

Brass protractors shall be flat and shall be between 0.031 and 0.062 inch thick except that 4.25-inch protractors shall be more than 0.015-inch, but less than 0.062 thick. The horizontal axis shall be along either edge of the crossbar. A vertical line shall be formed on the crossbar to indicate the vertical axis of the protractor. The line shall be formed by the same method as that used for the graduations.

Plastic protractors shall be flat and shall be between 0.062 and 0.125-inch thick. Graduations shall be on the bottom surface of the plastic and shall be readily distinguishable through the thickness of plastic. Letters and numbers may be on either surface of the protractor but shall be in reading position when the graduated surface is placed in contact with the drawing, map, or other work being performed.

The outer edge of the arc shall be graduated in degrees so that the zero and 180 degree marks are on the horizontal axis of the protractor. When the upper edge of the crossbar forms the horizontal axis, the graduations may be extended below that axis. The graduation marks that extend below the zero and 180 degree marks of the horizontal axis are the arc extensions.

Triangle. This commercial item description covers type I transparent plastic triangles, class A (30 - 60 degree angle, size 6-inch) and class B (45 degree, 4-inch). The triangles shall be plastic with the thickness specified in table II. The angles shall be within ±2

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minutes of the angle specified. The outside or ruling edges shall be straight and true within +0.003 inch. The surfaces of the triangle shall be flat and true within 0.010 inch.

TABLE II. Triangle Thickness.

Size (inches)	Thickness (inch) 1/
4	0.05 to 0.09
6	.05 to .09
8	.05 to .09
10	.07 to .11
12	.07 to .11
14	.07 to .11
16	.09 to .13
18	.09 to .13

1/ These are thickness limitations, not commercial standard thicknesses.

Contour ruling pen. The contour pen shall consist of a pen blade assembly attached to a thin rod which rotates in a tubular handle. The pen shall be constructed so that the blades are offset to the centerline of the handle and swivel without binding. A locking device shall hold the blades in a fixed position and a knurled screw shall permit adjustment of the blade points. The pen shall conform to the size and design shown in figure 1.

Railroad ruling pen. The pen shall consist of two pairs of pen handles with a tubular handle constructed so that the pairs of pen blades shall turn freely or remain fixed on the handle as desired. The pen blades shall be curved so that the points are eccentric to the center line of the handle, similar to the design of a contour pen in order that the pen blades will readily follow sharp curves. The overall length of the pen, including the handle shall be 5.25 inches. The pen shall be designed to provide lightness, good balance, and ease in handling.

Curve. Curves shall be furnished by pattern numbers 1 or 2 (see figure 2), in sets or individually as specified. The size of the curves shall be the size regularly produced by manufacturers of this type of curve. Curves shall be methacrylic plastic or acrylic plastic.

Large friction compass. The large friction compass shall be between 6 and 6.5 inches long, and shall permit drawing circles from 0.062 to not less than 5 inches in diameter. The friction shall be sufficient to hold the dividers in any position during normal use. One leg of the compass shall be provided with a removable and

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adjustable needle and the other leg shall have an interchangeable pen holder and pencil holder. The pen holder or the pencil holder shall be held firmly in place without movement when tightened. The pen holder portion shall have pen blades conforming to ruling pens, above. The pencil portion shall accommodate leads conforming to pencil leads, above.

Lengthening bar. A lengthening bar or beam not less than 3.5 inches long shall be provided for use with the large compass to permit drawing larger circles than are normally possible with the compass alone. The lengthening bar or beam shall fit into the socket for the pen or pencil attachment, and the pen or pencil attachment shall fit into the lengthening bar or beam.

Small divider. The small divider shall be similar to the small bow compass specified in this document, except that both legs shall have adjustable and replaceable needle points as specified herein.

Large divider. The large divider shall be friction type and shall be between 6 and 6.5 inches long. Both legs shall be equipped with adjustable and replaceable needles as specified in needle points cited herein. The range, or spacing between the points shall be continuously variable from less than 0.062 inch to more than 2.5 inches. Movement shall be free without binding or tight spots. The friction shall be sufficient to hold the dividers in any position during normal use.

Center horn. The center horn shall be a flat disc of transparent plastic mounted in a metal or plastic rim. The plastic disc shall be between 0.50 and 0.625 inch in diameter. The rim shall be equipped with three equally spaced needle-sharp pins protruding between 0.046 and 0.078 inch below the lower surface of the metal rim.

Additional parts and pencil lead holders. Additional parts and pencil lead holders may be in two separate holders or they may be combined into a single holder. Each holder shall be provided with a friction or threaded type removable cap. The holder shall contain not less than the following items:

3 Pencil leads.
3 Needle points.
2 Adjusting T-bolts and nuts (when required).
2 Eye bolts and clamp nuts (when required).

Screwdriver. When required for making adjustment on any instrument in any set, a screwdriver of the proper size shall be furnished with the set.

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Regulatory requirements: In accordance with Federal Acquisition Regulation, section 23.403, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

Certification of commercial item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Commercial Item Description are met.

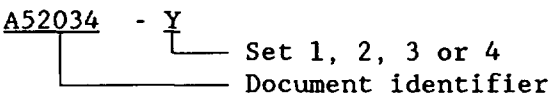
If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting office has the option of accepting or rejecting the product.

Preservation, packing, and marking. The preservation, packing, and marking shall be as specified in the contract or order.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

Part or identification number (PIN). The part or identifying number for drafting instrument sets and drafting instruments shall be as follows:



Notes:

1. Title, number, and date of this item description.

2. Purchaser should specify set number.

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3. When this commercial item description is used for procurement, the commercial item certification clause must appear in the solicitation.

4. FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

5. ASTM B 36, ASTM B 211, and ASTM B 455 are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

6. Preparing activity for this CID is: Commander, US Army Belvoir RD&E Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

MILITARY INTERESTS:

Custodians
Army - ME
Air Force - 99

Review activity
DLA-GS

User activities
Navy - MC, YD

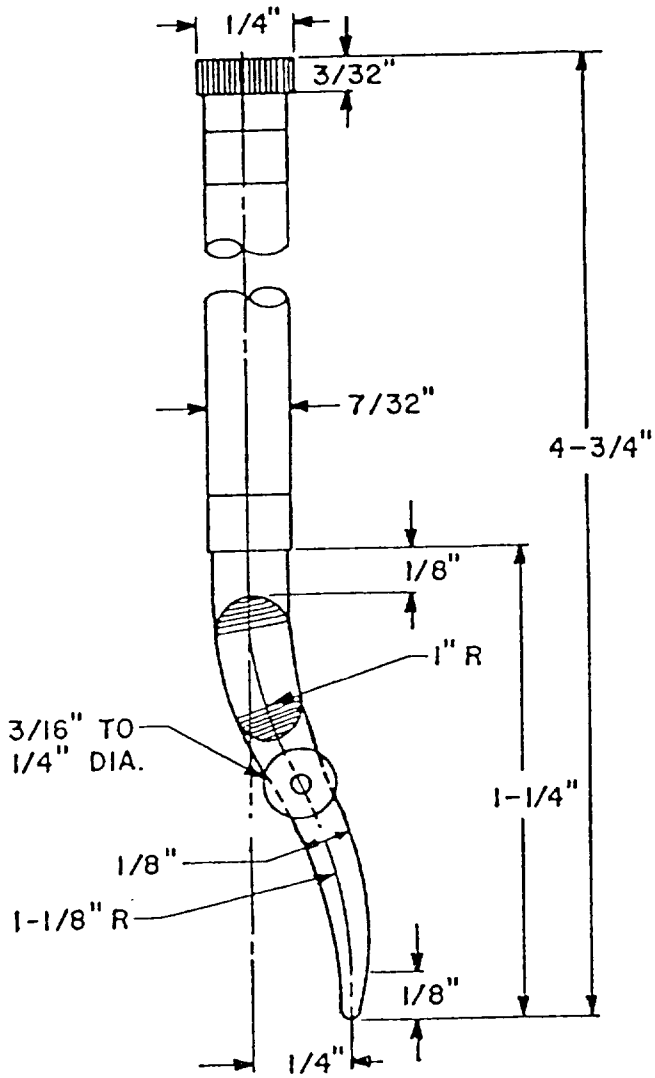
CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
VA-OSS

Preparing activity:
Army - ME

Project 6675-0027

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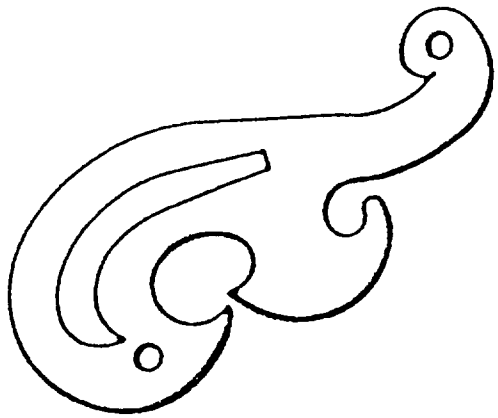


NOTE:
DECIMAL DIMENSIONS TO BE HELD WITHIN ±0.01".
FRACTIONAL DIMENSIONS TO BE HELD WITHIN ±1/32"
UNLESS OTHERWISE SPECIFIED.

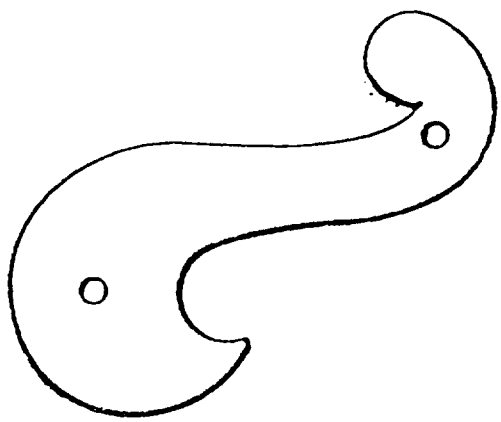
FIGURE 1. Contour Pen

X-2306

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Pattern 1



Pattern 2

FIGURE 2. Curves

X-5016

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